Explainable, Anonymized Resume Shortlisting System

A decision-support tool that ranks resumes against a job profile and leaves interviews and the final decision with HR.

1. Goal

Build a tool that reads the resumes a company receives, compares them with a job profile, and gives HR a ranked shortlist with clear reasons for every ranking. HR keeps the interview round and the final hiring decision.

Used for scoring	Education (degree level and field), work experience, projects, skills
Never used	College name, candidate name, gender, age, photo, address

2. Success targets

- Top-10 agreement with recruiter shortlists of 80% or more on a labeled test set (replace with real results after testing).
- Identical scores when college name and personal details are changed (the bias test passes).
- 100 resumes screened in a few minutes, with a written reason for every ranking.

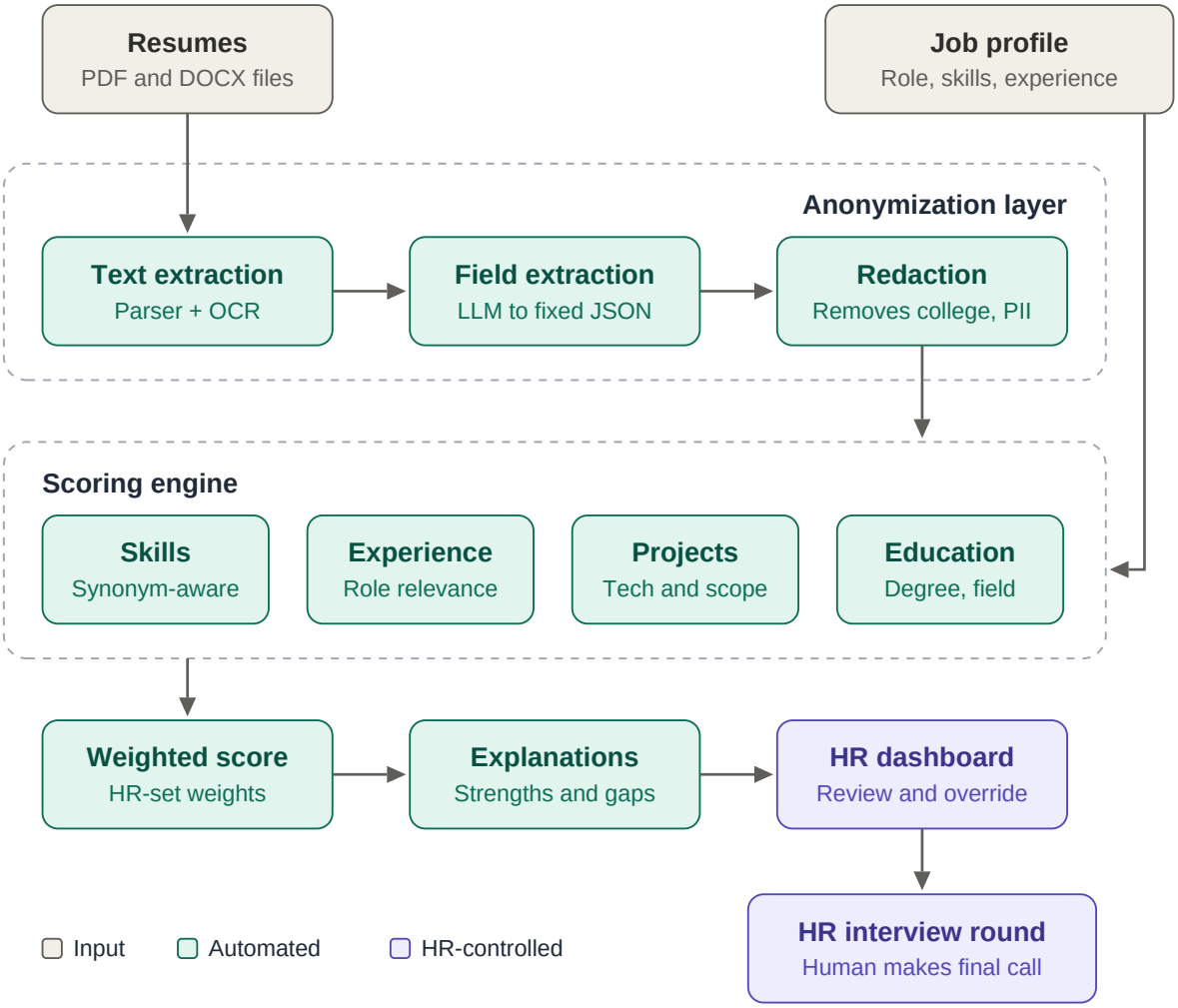
3. Design principles

- **Human in control:** the system flags and ranks but never auto-rejects.
- **Explainable:** every score comes with a written reason.
- **Anonymized:** the scoring layer never sees identifying fields.
- **Adjustable:** each company sets its own weights.
- **Private:** resumes are deleted after a set retention period.

4. Competitor positioning

Tools such as HireVue automate the interview stage. This system works only at the shortlisting stage and keeps human judgment in the HR round. Compared with large ATS platforms, it aims to be a lightweight, transparent, adjustable screener for small and mid-size teams.

5. System architecture



- Resumes pass through the **anonymization layer** first, so the scoring engine never sees college name or personal details.
- The **job profile** goes straight to the scoring engine, where each resume is scored on skills, experience, projects and education.
- The weighted score and written explanations reach the **HR dashboard**, where HR can review and override. The interview round and final decision stay with HR.

6. Tech stack

Layer	Tools
Backend	Python, FastAPI
Parsing	pdfplumber / PyMuPDF, python-docx, Tesseract (OCR)
NLP / AI	spaCy, sentence-transformers, LLM API for extraction and explanations
Storage	PostgreSQL, FAISS / Chroma or pgvector
Frontend	Streamlit (MVP), later React
Deployment	Docker, then Render / AWS / GCP
Testing	pytest, GitHub Actions

7. Project phases

Phase 1: Foundation and MVP — Weeks 1-2

Aim: Upload a job profile and resumes, get a basic ranked list.

1.1	**Setup**	Create the repo, folder structure, virtual environment and README skeleton. Set up Git workflow and basic CI.
1.2	**Resume ingestion**	Upload endpoint for PDF and DOCX, text extraction for each format, OCR fallback for scanned resumes.
1.3	**Job profile input**	Form for role title, required and nice-to-have skills, minimum experience and education requirement; stored in the database.
1.4	**Baseline ranking**	Generate embeddings for the job profile and each resume; rank by cosine similarity.
1.5	**Basic UI**	Streamlit page to upload, run and view the ranked table.

Deliverable: working end-to-end demo with a simple ranked list.

Phase 2: Anonymization and structured scoring — Weeks 3-4

Aim: Make scoring fair, structured and multi-factor.

2.1	**Structured extraction**	Convert each resume to fixed JSON: degree level, field of study, experience years, roles, projects, skills, certifications. Validate and handle missing fields.
2.2	**Redaction layer**	Remove college name, candidate name, gender, age, photo, address and graduation year. Redact college mentions inside project descriptions. Score roles and responsibilities instead of employer brand.
2.3	**Multi-factor scoring**	Skill match with synonym handling, experience fit, project relevance, education fit (degree and field only), weighted final score.
2.4	**Adjustable weights**	Sliders or config so each company sets its priorities. Hard requirements flag candidates rather than reject them.
2.5	**Bias test**	Run each resume with and without college name and personal details and confirm identical scores; add as an automated test.

Deliverable: anonymized, multi-factor scoring with a passing bias test.

Phase 3: Explainability and HR workflow — Weeks 5-6

Aim: Make it useful for real recruiters and keep HR in control.

3.1	**Explanations**	Per-candidate summary of strengths, gaps and why the score is what it is.
3.2	**HR dashboard**	Ranked shortlist with score breakdown by factor, side-by-side comparison, filters by skill, experience range and score.
3.3	**HR control features**	Manual override with notes, a 'review manually' flag for borderline candidates, full resume revealed only after shortlisting.
3.4	**Interview support**	Suggested interview questions based on each candidate's gaps and projects.
3.5	**Export**	CSV and PDF shortlist report.

Deliverable: a recruiter-ready dashboard where HR can review, override and export.

Phase 4: Scale, evaluation and deployment — Weeks 7-8

Aim: Make it production-like and prove it works with numbers.

4.1	**Accuracy evaluation**	Label 50-200 resumes for 2-3 job profiles; measure precision@k, top-10 agreement and ranking correlation; compare with a keyword-only baseline.
4.2	**Scalability**	Batch processing for hundreds of resumes with a queue (Celery + Redis), embedding cache and progress tracking.
4.3	**Multi-company support**	Authentication, separate workspaces per company, role-based access (HR admin, recruiter).
4.4	**Privacy and security**	Automatic deletion after a retention period, encrypted storage, no resume content in logs, consent and data-handling notes.
4.5	**Deployment and docs**	Dockerize and deploy with a live link; README with architecture diagram, demo GIF, competitor comparison, bias-mitigation section and results; unit tests and CI.

Deliverable: deployed product with measured accuracy and complete documentation.

8. Timeline summary

Phase	Weeks	Main outcome
1. Foundation and MVP	1-2	Upload, parse, rank
2. Anonymization and scoring	3-4	Fair, multi-factor scoring
3. Explainability and HR workflow	5-6	Dashboard, overrides, interview questions
4. Scale, evaluation, deployment	7-8	Live app with measured results

9. Resume bullet

Built and deployed an explainable, anonymized resume-shortlisting platform (FastAPI, sentence-transformers, LLM) that ranks candidates on education, experience, projects and skills without using college name or personal identifiers; achieved [X]% top-10 agreement with recruiter shortlists on [N] resumes while keeping interviews and final decisions with HR. (Fill in X and N from your Phase 4 evaluation.)